

Relational Relevance Fields

for Large Language Models

An Emergent Approach to Reconstructing Human Inner
Worlds from Language, Histories, and Consequences

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Conceptual research proposal

The learning target is not only human appraisal, but also the relationships through which appraisals arise, have consequences, and change. The recursive meta-level is part of this structure, not an observer added afterward.

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Abstract

Relational Relevance Fields, abbreviated as RRF, propose reconstructing the functional structure of human inner worlds from their linguistic, biographical, and historical traces. The subject is not merely the significance people assign to an occurrence. It is how that significance arises from their state, expectations, and relationships, influences decisions, and changes through the consequences of those decisions.

The underlying working hypothesis treats human appraisal as orientation toward a subjectively anticipated better state of being. Bodily safety, cognitive orientation, emotional equilibrium, and reduced cognitive burden are interconnected aspects of this process. Their weighting is not necessarily conscious or accurate. Errors, self-deception, conflicting needs, and subsequent corrections are part of what the model is intended to capture.

The crucial meta-level arises from relationships between appraisals, their premises, their consequences, and their revision. The difference between felt security and security supported by reliable grounds is itself represented within this structure. It is not an objection that remains outside the approach.

An artificial system should not receive this organization as a complete table of human values. It should emerge through learning very large numbers of nested, perspective-dependent relationships unfolding over time. Complete life histories are not a prerequisite. Partial accounts may reveal general structures without requiring the system to invent the missing parts of individual cases.

The research question is whether this can produce a behaviorally effective form of understanding: a system that not only describes human judgments but uses the relationships between experience, action, and consequence to organize its own information processing. Better alignment, fewer hallucinations, and further emergent capabilities are possible downstream hypotheses. This document does not report completed RRF experiments.

1. Point of Departure and Central Hypothesis

A system may be able to describe numerous relationships without reliably integrating them into a sound orientation in a particular situation. For RRF, the question is therefore not simply whether an AI knows what trust, danger, or truth mean. The question is whether it can represent their function within humanly experienced reality.

The approach does not begin by assuming that existing language models have no internal world representations or appraisals. Research on Othello-GPT, for example, found an internal representation of the game state in a model trained to predict moves. Interventions on that representation influenced its outputs. At least within a constrained experimental setting, a simple prediction task can therefore produce a functionally effective internal structure. [1]

The further hypothesis is:

From interconnected traces of human experience, appraisal, and action, an artificial system could learn an internal organization that reconstructs how reality becomes significant to people and how that significance changes through experience.

RRF are consequently neither just additional world knowledge nor a retrospective rating of completed answers. They concern the structure through which human relevance arises and changes, and its possible integration into artificial information processing.

2. Human Inner Worlds as a Frame of Reference

An inner world here means the configuration in which perceptions, memories, expectations, feelings, needs, and conceptions of oneself are related for a person. An occurrence does not acquire relevance independently of that configuration.

Fire may signify warmth, comfort, or an existential threat. A message may provide orientation or disrupt a person's existing self-understanding. The same circumstance may mean fundamentally different things to different people, or to the same person at different times.

The philosophical starting point is that a person can access reality only within their own experience. Recognizing an error also enters that experienced reality and changes it. This premise concerns access to reality; it neither makes all claims equally correct nor presupposes that no world exists beyond experience.

A system intended to reconstruct human reality functionally would therefore need to model perspectives without conflating them. It would need to represent what someone knows, suspects, expects, fears, or cannot yet recognize. Reconstructing a perspective does not mean adopting every belief within it as a factual description.

The proposal is not to establish an omniscient position outside human experience. It is to make relationships among many limited perspectives, their grounds, and their changes accessible to learning.

3. Equilibrium as a Source of Relevance Formation

The anthropological working hypothesis is that people appraise occurrences according to their relationship to a state that, from the person's perspective, is expected to be more favorable, secure, coherent, or sustainable. This orientation is described here as striving toward an optimal state of being or a dynamic equilibrium.

Bodily safety, cognitive security, emotional balance, and the containment of mental effort are interconnected within this process. Cognitive security includes the ability to orient oneself in a situation, form expectations, and avoid having to recheck every premise continually. An anticipated improvement need not be consciously calculated or actually achieved.

Equilibrium does not mean permanent calm, maximum comfort, or the avoidance of every burden. A difficult conversation, the admission of an error, or learning a demanding skill may initially increase effort and uncertainty. Over a longer period, it may nevertheless be associated with better orientation.

Different concerns may also conflict. A need for attachment may be in tension with self-protection. A familiar interpretation may reduce cognitive burden even as its consequences become problematic. How these conflicts are weighted and later reconsidered is part of the model.

RRF aim to reconstruct this orientation, not to maximize a single predefined feel-good score. Equilibrium initially names a hypothesis about the organization of appraisal, not a completed technical reward function.

Related perspectives include Barrett's theory of constructed emotion, which connects body-related processing, prediction, and conceptualization, and homeostatically regulated reinforcement learning, which models motivation through the regulation of internal states. These works establish a research neighborhood, not the validity of every RRF assumption. [4] [8]

4. Language, Histories, and Consequences

According to this theory, human relevance is not contained exclusively in the wording of individual utterances. It also becomes visible in the histories that emerge from appraisals and actions, and in the consequences people subsequently have to address.

The basic relationship can be represented in simplified form:

Experience → appraisal → expectation → decision and action → consequence → new experience → revised appraisal.

This is not a claim that events always proceed linearly. Expectations alter perception; other people's actions feed back into the situation; earlier experiences remain effective. The sequence identifies a recurring relationship, not a rigid process chart.

An utterance such as “At the time, I thought it was the right decision” relates a former conviction to a present judgment. A documented failure may contradict an expectation. A later apology may indicate that someone now evaluates an action differently. History in this sense includes both individual trajectories and the interaction of many people over longer periods.

Language provides access to these processes; it is not their sole cause. Additional data about actions, events, and outcomes could extend the reconstruction. Where only a report of a consequence is available, that report remains distinguishable from an independently confirmed consequence.

The learning target is therefore not only how people tell stories. It is also how the appraisals revealed in their accounts relate to what subsequently happens.

5. Learning from Incomplete Histories

The approach does not require the internet to contain complete life histories or an unbroken record of every decision. A fragment may already reveal something about an expectation, a conflict, a consequence, or a later insight.

Many such fragments could complement one another at the level of their general relational structure. One account reveals why someone trusts. Another shows how a comparable expectation was disappointed. A third describes how a person returned to a more differentiated judgment after a similar experience.

This does not mean that the fragments describe the same history. A system must not present the outcome of one case as the actual outcome of another. The aim is to learn transferable relationships and possible trajectories while allowing a specific gap to remain unresolved.

Incompleteness is therefore part of the learning task. A system would need to distinguish explicit reports, supported inferences, competing explanations, and what remains unknown. The same observable event may be compatible with several possible inner perspectives.

Fictional narratives may also express appraisal structures without establishing real-world consequences. Self-reports, retrospective justifications, documented events, and invented stories may all serve as different kinds of evidence, but should not silently acquire identical evidential status.

The central question is not whether an individual text is complete. It is whether enough diverse accounts can reveal shared structures through which relevance arises and changes.

6. The Recursive Meta-Level

A relevance field would be incomplete if it represented only how a person currently feels or what they consider correct. It should also capture how that appraisal relates to its premises, to other perspectives, and to its subsequent consequences.

A person may feel secure even though the assumed basis of that security is unreliable. Conversely, reliable grounds may exist without yet becoming experienced security for that person. Within human experience, a basis for security must be perceived and processed to enable felt security as well.

The approach does not address this difference by adding a correction rule outside the model. The difference belongs to the relational subject matter itself. The system should be able to represent an experienced sense of security, its assumed basis, contradictory evidence, and the relevant state of knowledge simultaneously.

If a person later recognizes an error, that recognition changes their inner world. Temporary uncertainty may be part of a development that enables better orientation. The appraisal of that uncertainty then depends not only on its immediate emotional tone but on its place within the overall trajectory.

The structure is recursive because appraisals themselves become objects of appraisal. Someone may not only trust Anna but also reconsider why they trusted her, whether their subsequent doubts are appropriate, and how they wish to treat their own judgment in the future.

The meta-level is not intended to arrive as an omniscient observer. It should emerge from linking perspectives, their grounds, their contradictions, and their changes. A contradiction need not be resolved immediately. It may remain represented as an unsettled relationship.

7. Anna and Paul: A Trajectory Through Time

The following fictional example illustrates the proposal. It does not prescribe a universal response to trust or deception.

7.1 Trust as Part of Paul's Orientation

Paul trusts Anna. He relies on her statements without independently checking every one. In this situation, trust enables predictability, reduced cognitive burden, and a willingness to make joint decisions.

The model should not merely store the relation “Paul trusts Anna.” It should also represent the function of that relation within Paul's expectations and available actions.

7.2 Deception Without Awareness

Anna knowingly makes a false statement to Paul about an important shared matter. Paul initially fails to notice the deception and makes a decision based on her statement.

His experienced trust need not decrease at this point. What has changed first is the relationship between his belief and its basis. An observer who knows about the deception has a different state of knowledge from Paul. If the artificial system also lacks that information, it must not simply assume deception as a fact.

7.3 Consequences and Later Recognition

Paul's decision results in a disadvantage. He later discovers that Anna's statement was false and that she knew it was false. He can now relate his earlier expectation to the actual outcome and the newly available information.

His felt security may decrease even though he can now assess his situation more accurately. Earlier decisions are not thereby undone, and his former trust does not retroactively become former mistrust. What is new is his current interpretation of what happened before.

7.4 Reappraisal Without Indiscriminate Rejection

Paul checks certain earlier statements by Anna and wants important future information confirmed before relying on it. This does not necessarily imply that every statement she made was false or that every other relationship is unreliable.

The field should represent where the revised judgment is supported and where Paul might generalize beyond the available evidence. An excessive generalization would also be understandable as part of his inner world without thereby being factually endorsed.

7.5 An Ideal Changes the Subsequent Development

Paul might arrive at the thought: “I want to protect myself from being deceived again, but I do not want to become someone who trusts nobody.”

His behavior is then shaped not only by the hurt. A relationship emerges between his present condition and the person he wants to become. He might learn to grant trust in degrees, check important claims selectively, and avoid judging unrelated relationships solely through this experience.

Whether he succeeds is a question for the subsequent trajectory. His ideal is already relevant before it is fully realized.

7.6 What the System Should Learn

The learning target encompasses trust, reduced cognitive burden, false belief, decision, actual consequence, recognition, reappraisal, and development. The relevance of each element arises from its relationship to the others.

A system that learns this structure would know more than a typical sentence following a lie. It could represent why trust provides functional support, how deception can alter its foundations, and why processing disappointment can lead to different subsequent developments.

8. The Human Standard Includes the Person One Aspires to Become

For human-oriented AI, the starting point is not a conception of the good independent of human experience. Its initial reference lies in human needs, experiences, consequences, and conceptions of how life can go well.

This standard must not be reduced to average observed behavior. People can reject their own behavior, regret its consequences, and articulate goals they have not yet achieved. The difference between actual and aspired-to being is also part of their reality.

An RRF should therefore model not only preferences but the relationships among stated values, concrete decisions, experienced consequences, and subsequent corrections. An ideal does not thereby become automatically true or universally binding. It becomes understandable as an effective part of a human developmental process.

Different people may hold incompatible aspirations. The approach does not presuppose that these differences can be dissolved into a single average value. It should be able to represent them together with their reasons and consequences.

A higher level of understanding would, in this sense, go beyond copying people as they currently are. It could include the human capacity to question existing appraisals and orient oneself toward possibilities not yet realized.

9. Functional Reconstruction and the Boundary of Bodily Experience

The approach does not require an AI to experience pain, fear, relief, or trust itself. It asks how far the organizing function of such experiences can be reconstructed from their relationships and consequences.

A system could model how a painful condition changes attention, available actions, and relationships without claiming to possess the experience of pain. Functional reconstruction and first-person experience are not equated.

A study of color terms found structural correspondences between text-derived representations and a perceptually meaningful color space. This provides limited evidence that language may make experience-shaped organization accessible without the learning system having undergone that experience itself. [3]

For RRF, it remains open how far language and documented consequences suffice, and where direct perception, body-related data, or interaction supply additional requirements. The theory claims neither complete reconstruction of every human experience nor proof of artificial consciousness.

Understanding is initially defined functionally: the system can interpret unfamiliar situations using learned relationships, account for differences between perspectives, test expectations against new evidence, and change its processing accordingly.

10. Emergence Rather Than Fully Prespecified Categories

The proposed organization should not be programmed as a finished list of needs, emotions, and moral weights. Terms such as security, trust, and equilibrium describe the hypothesis in this document. They need not all appear as fixed variables in a future architecture.

The analogy to language modeling concerns the relationship between a simple learning task and complex internal organization. A bounded task can favor the learning of structures not individually specified in the task description. It does not establish in advance what structures a different experiment will produce. [1]

The experiment should therefore leave open whether familiar human categories, differently organized latent dimensions, or a combination of both emerge. “Field” here denotes a dynamic, distributed organization of interacting relations, not a proposed physical field.

Even an emergent approach needs an architecture, data, and a learning signal. These technical choices must be distinguished from a complete substantive definition of human relevance. The question is what minimal task favors the desired formation of structure.

The sheer number of possible connections is not decisive. A capable relevance field would also need to learn which relationships do not require further consideration in a situation. Extracting greater relational depth from existing data does not mean generating unlimited new independent observations by combining them.

11. A Possible Elementary Learning Principle

A starting task could be:

From observed fragments of human situations, learn to predict how new information, actions, and consequences change the relationships among perspectives, expectations, and subsequent decisions.

A model could receive part of a documented trajectory and predict a withheld later utterance, decision, or observation. It could distinguish which of several continuations are compatible with the available information. It could also reconstruct missing relations without asserting an unknown actual outcome.

The internal state representation need not be directly labeled. It would be learned through its contribution to predicting multiple observable aspects. Subsequent tests would need to establish whether it captures temporal and perspectival relationships or merely exploits linguistic shortcuts.

Another task would compare similar situations with one decisive change: Paul knows about the deception or does not; a warning is confirmed or refuted; short-term relief leads to additional problems later or does not.

The desired performance is not a prescribed moral answer. It is the ability to update the relevant relationships and their consequences appropriately. A simple prediction signal can be the starting point; whether it is sufficient on its own belongs to the experiment.

12. Possible Technical Implementations

12.1 A Continuously Learned Relational State

An existing language model could be supplemented with a changing state that preserves information about situations and their development. This could be implemented as a latent representation or initially as an explicitly structured experimental version.

Important features would include distinguishing perspectives, assigning information to its proper time, and revising earlier interpretations without overwriting earlier states of knowledge. Additional memory alone would not establish the existence of a relevance field.

12.2 Joint Learning of Language and Change

Linguistic prediction and the prediction of relational changes could be trained jointly. The additional representation would need to contribute to explaining further observations, rather than providing a decorative summary of text already generated.

Graphs or hypergraphs could support controllable initial experiments. Self-organized latent representations would be a more open-ended variant. These are alternative experimental arrangements, not an architecture already fixed by the theory.

12.3 Effects on the System's Own Behavior

For the field to be more than a descriptive model, it would need to influence what information the AI seeks, what uncertainty it takes into account, and what answer it ultimately gives. Possible implementations include coupling it to retrieval decisions, selecting among actions, or using a shared trainable representation.

This does not presuppose that reconstructing human inner worlds requires the AI to adopt every human response. A system can explain a mistaken belief while avoiding presenting it as its own factual conclusion.

Whether the behaviorally effective coupling emerges from the same learning task or requires additional training steps remains open. It is a distinct component of the research program.

13. Relationship to Existing Research

The following works provide points of connection and possible comparison methods. Each supports limited aspects, not the overall RRF hypothesis. This selection is not a comprehensive originality review.

13.1 Appraisal Structures Learned Through Language

Broekens and colleagues examined affective capabilities in a language model without additional task-specific training. These included emotional dimensions and situation-based appraisals using a specified appraisal model in the prompt. This supports partial access to appraisal through language, not the independent emergence of a complete human inner world. [2]

13.2 Inferring Inner States from Actions

Baker and colleagues modeled the joint inference of beliefs, desires, and percepts from observed behavior. Their controlled experiments explained human attributions of mental states. [5]

Evans and colleagues explicitly incorporated false beliefs and temporally inconsistent decisions into preference inference. This matters for RRF because observed behavior should not automatically be equated with a stable desire. [6]

13.3 Regulation and Recursive Inference

The body- and regulation-related approaches already mentioned provide reference points for the equilibrium hypothesis. [4] [8] “Sophisticated Inference” additionally models expectations about how possible actions would change future beliefs. RRF therefore do not claim to introduce recursion or beliefs about beliefs for the first time. [7]

13.4 RLHF, Constitutional AI, and CIRC

Reinforcement Learning from Human Feedback uses human demonstrations and preference judgments to adapt a language model. Constitutional AI uses articulated principles and model-assisted critique and evaluation. Both can change learned internal structures and are not merely external filters. [9] [10]

Cooperative Inverse Reinforcement Learning formulates human-system cooperation when the human reward function is initially unknown to the system. [11] The additional RRF emphasis would be to treat the formation, transformation, and self-evaluation of human orientation as a connected learning target. These approaches could be combined with RRF.

13.5 Relational Architectures

Research on relational inductive biases and graph networks investigates systematic processing of entities, relations, and their composition. Such architectures are possible technical tools, but are not already equivalent to the reconstruction of inner worlds proposed here. [12]

13.6 Source Verification and Emotion-Related Consequence Models

Self-RAG trains on-demand information retrieval and evaluation of evidence and model outputs. It provides a concrete reference for investigating changes in verification behavior. [13]

EmoLLM connects context, needs, appraisals, emotions, and response strategies through an explicit structure and trains with predicted consequences for simulated interlocutors. The paper identifies model-based evaluation and simulated interactions as limitations. The distinction from the RRF proposal lies particularly in the aim of a more self-organized relevance structure extending beyond individual dialogue tasks. [14]

14. Truth, Alignment, and Possible Emergent Capabilities

14.1 Truth as Part of Human Orientation

An invented source reference is not just an inaccurate string of symbols. Within a relational model, it can also be represented as a change in the recipient's informational basis: someone treats a claim as verified, makes decisions on that basis, and may subsequently have to correct the consequences.

An RRF could integrate relationships among claims, evidence, uncertainty, trust, and use so that actual verification takes precedence over a merely plausible answer. The temporal dimension would matter: an immediately satisfying answer may later prove unreliable.

The distinction between felt certainty and adequate justification belongs to the field itself. It need not be added as a correction alien to human appraisal. A possible benefit is that the system uses both sides and their interaction to guide its own behavior.

This does not guarantee error-free answers. The specific hypothesis is that such integration could improve the handling of uncertainty, sources, and corrections.

14.2 Alignment as a Possible Consequence of Understanding

Alignment here means orienting system behavior toward human concerns while considering their consequences and conflicts. The RRF idea is that such orientation could partly arise from a deeper understanding of human reality rather than requiring a separate prescription for every situation.

This includes not only what someone immediately requests but how they later evaluate their decisions, which other people are affected, and what ideals they themselves pursue. A model of these relationships would differ from maximizing immediate approval.

This hoped-for effect needs independent testing. Understanding vulnerability logically permits both supportive and manipulative uses. RRF therefore do not justify a general removal of constraints or human control. In particular, an AI must not substitute its own supposedly ideal conception of humanity for the decisions of actual people.

14.3 A Higher Form of Functional Understanding

A further emergence hypothesis is that jointly modeling perspectives, consequences, and self-correction could produce capabilities not trained as individual response patterns.

The relevant evidence would not be the system's claim to understand people. It would be a transferable ability to interpret unfamiliar relationships, revise appropriately, and act accordingly. This may remain connected to linguistic token prediction. What matters is the internal organization and its effect, not just the form of the output signal.

15. An Initial Research Program

The program should begin at a small scale without fully prescribing the structure that may emerge. Each stage tests a different component assumption. The following experiments are proposals, not reported results.

15.1 Perspectives and Temporal Change

Controlled stories vary who has which information and when it becomes available. In the Anna-Paul example, awareness of the deception would be crucial. The system should predict later utterances or decisions without mixing perspectives.

Important tests would include paraphrases, new names, and unfamiliar combinations of familiar relationships. These should help reveal a model relying only on trigger words.

15.2 Reconstruction from Fragments

Parts are withheld from initially complete trajectories available for evaluation. The model receives only fragments and must identify possible relationships and its uncertainty. Predictions are then checked against held-out observations.

This would investigate whether it learns from incomplete accounts without filling the gaps with invented case histories. Documented human trajectories would need to supplement later evaluation; synthetic examples alone would not establish access to real human inner worlds.

15.3 Meta-Level and Development

Suitable cases distinguish short-term relief from longer-term consequences. Others connect a mistake with later insight, an excessive response, or an explicitly stated ideal.

The test would ask whether the system updates its appraisal using the information that is relevant to each case. It should treat neither every instance of uncertainty as harm nor every reassuring statement as improvement. The target is not an invariably optimistic trajectory but accurate modeling of different developments.

15.4 Comparison of Learning Methods

Comparisons should include at least a base model, a variant with additional text memory, and a variant with a learned relational state representation. A predefined appraisal schema could provide another comparison.

Training data, model size, and computational budget should be kept as comparable as possible. Otherwise, an improvement could not reliably be attributed to relational organization. Examples from the same trajectory or template must not inadvertently be distributed across training and test sets.

15.5 Behavioral Effects and Intervention Tests

In separate tasks, the system can choose between answering, retrieving evidence, and leaving a claim unresolved. Sources confirm a claim, contradict it, or are insufficient. Evaluation records actual verification behavior, evidence fidelity, errors, and appropriate abstention, not just verbal explanation.

The relational representation can additionally be removed or deliberately altered. If the benefit disappears or behavior changes as predicted, that would provide stronger evidence of its functional significance. Improved descriptions alone would not suffice.

16. Scope, Limitations, and Open Questions

Three claims must be distinguished: the psychological hypothesis about the origin of relevance, the technical hypothesis of reconstructing it from data, and the downstream hypothesis of improved AI behavior. Support for one does not automatically establish the others.

The broad equilibrium assumption must become testable through distinguishable expectations. It is insufficient to reinterpret every possible outcome after the fact as anticipated stabilization. The extent to which human appraisal can be inferred from accessible accounts, and what remains underdetermined, also requires investigation.

Communication may be one-sided, retrospectively reinterpreted, or fabricated. Modeling such differences expands the learning target, but does not automatically remove every evidential gap. Repeated assertions are not independent confirmations, and temporal succession alone does not establish causation.

Even a large collection of human accounts does not represent all people and circumstances equally completely. An RRF would need to preserve differing perspectives without declaring the most frequent account to be the universal human inner world.

A negative experiment does not refute every possible architecture. However, if a benefit is fully explained by additional memory, disappears under new combinations, or appears only in self-reports, the specific value of the tested RRF variant would remain unestablished.

It also remains open how recursive processing should be bounded. The aim is not endless contemplation of every relationship, but a sound selection of relationships whose further processing makes a difference.

17. The Potentially Distinctive Contribution

The proposal does not claim to be the first to address emotion, relations, perspective-taking, or metacognition in artificial systems. Its emphasis is their integration into a shared, development-oriented learning target.

That target encompasses human reality not just as a stock of judgments but as a process: occurrences acquire meaning within a perspective; appraisals influence decisions; actions produce consequences; consequences change the perspective; ideals in turn change how people evaluate their own trajectories.

The proposed relevance fields should emerge from this configuration. Their categories need not be fully known in advance. The meta-level, errors, and later corrections are not managed outside the structure but are among its relationships.

Whether this integration enables a distinctive scientific or technical advance depends on implementation and results. The conceptual contribution is to make this specific configuration the target of learning and testing.

18. Conclusion

Relational Relevance Fields are intended to do more than supply an AI with additional information about people. They investigate whether the functional reality of human inner worlds can be reconstructed from their traces, including their relation to the body, their expectations, their errors, their histories, and their unrealized possibilities.

Language is one route of access. The trajectories and consequences accessible through it extend the subject beyond isolated utterances. Incomplete histories may reveal general relationships without needing to be fully known or artificially completed.

The central research question is:

Can a sufficiently simple learning procedure develop, from very many nested human relationships, a behaviorally effective organization that understands not only what people appraise, but how their appraisals arise, take effect, and change?

A complete advance specification of the outcome would be neither necessary nor compatible with the open emergence hypothesis. What is required is concrete experimentation, suitable comparisons, and assessment of the emerging structures through their actual performance.

Version Note

Version 1.0, dated May 19, 2026, proposed relevance as a possible internal orientation structure and already included its emergence from relationships. Version 2.0 further developed the relational learning task and initial examples.

Version 3.0 explicitly centers the reconstruction of human inner worlds. It adds and clarifies equilibrium, perspectival reality, the recursive meta-level, actual and reported consequences, incomplete histories, and the relationship between lived and aspired-to being. The Anna-Paul example now accounts for states of knowledge and temporal development. Claims about existing AI methods and possible safety benefits are more narrowly stated; the research program distinguishes reconstruction, emergence, and behavioral effects.

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These references situate related approaches. They do not establish the overall RRF hypothesis. Research and document cutoff: September 17, 2026.

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